

Bayesian data fusion developments for causal inference

Augmenting clinical trials with real-world observational evidence

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Paper & slides

The clinical trade-off:

- *Randomised controlled trials (RCTs)*:
 - Gold standard for internal validity through design;
 - Small sample sizes, high costs, restrictive criteria: limited statistical power and poor generalisability.
- *Observational real-world data (RWD)*:
 - Large patient cohorts, clinical registries, routine practice records;
 - Vulnerable to unmeasured confounding and selection bias.

Objective of data fusion

Borrow statistical power from large observational cohorts while preserving the internal validity of the randomized trial.

The Bayesian paradigm offers a natural framework to introduce auxiliary evidence into priors and adaptively discount it under data conflict.

Let $D_0 = (n_0, y_0, X_0)$ denote historical data with likelihood $L(\theta \mid D_0)$, and $\pi_0(\theta \mid c_0)$ an initial prior indexed by c_0 .

Power prior and posterior formulation

For a discounting hyperparameter $a_0 \in [0, 1]$:

$$\pi(\theta \mid D_0, a_0) \propto L(\theta \mid D_0)^{a_0} \pi_0(\theta \mid c_0), \quad \pi(\theta \mid D_e, D_0, a_0) \propto L(\theta \mid D_e) \pi(\theta \mid D_0, a_0)$$

- $a_0 = 1$: unpenalized pooling; $a_0 = 0$ excludes external data entirely.
- *Precision scaling*: Disperses historical prior concentration.

Limitations:

- *Fixed a_0* : Subjective elicitation; no internal commensurability check;
- *Random a_0* : Naive joint prior $\pi(\theta, a_0 \mid D_0) \propto L^{a_0} \pi_0 \pi(a_0)$ violates the Likelihood Principle (scaling L changes posterior on a_0) and ignores $C(a_0)$.

Proper power prior (Duan et al., 2006; Neuenschwander et al., 2009)

Retaining $C(a_0) = [\int_{\Theta} L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) d\theta]^{-1}$ restores probability calculus:

$$\pi(\theta, a_0 | D_0) = C(a_0) L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) \pi(a_0) \quad (a_0 \in A)$$

- *Prevents collapse*: Without $C(a_0)$, $\pi(a_0 | \text{data})$ collapses to 0 even for identical datasets.
- *Dynamic commensurability*: Given $\pi(a_0 | D_e, D_0) \propto \pi(a_0) m(D_e | D_0, a_0)$, concordance pushes $a_0 \rightarrow 1$; conflict forces $a_0 \rightarrow 0$.
- **Multi-study*: Extends to vector weights $a_0 = (a_{01}, \dots, a_{0H})^T$ for $H \geq 2$.

The single-source bottleneck ($H = 1$)

With only one historical study ($H = 1$), data cannot reliably separate between-study heterogeneity from noise: $\pi(a_0 | \text{data})$ remains sensitive to the chosen $\pi(a_0)$.

Schmidli et al. (2014): Shift to hierarchical predictive borrowing for control response $\theta = g(\psi)$ across $H \geq 3$ historical studies to downsize concurrent controls.

Hierarchical exchangeability model

$$\theta_1, \dots, \theta_H, \theta_* \mid \mu, \tau \stackrel{\text{i.i.d.}}{\sim} N(\mu, \tau^2), \quad \pi(\mu, \tau) = \pi(\mu)\pi(\tau)$$

where $\tau \geq 0$ quantifies between-trial heterogeneity (half- t or half-Normal).

- Small values of τ reflect cross-trial concordance and facilitate borrowing;
- *Conjugate mixture approximation:* The empirical MCMC predictive density is approximated via EM by a K -component mixture ($K \in \{2, 3\}$):

$$\hat{\pi}_{\text{MAP}}(\psi_*) = \sum_{k=1}^K w_k \pi_c(\psi_* \mid \xi_k), \quad \sum_{k=1}^K w_k = 1, \quad w_k > 0$$

Provides closed-form prospective updates, bypassing MCMC during trial analysis.

To protect against non-exchangeability, they construct a robust mixture:

Robust mixture prior (Schmidli et al. (2014))

Augment with vague conjugate component $\pi_V(\psi_*)$ weighted by $w_R \in [0.10, 0.20]$:

$$\hat{\pi}_{\text{HR}}(\psi_*) = (1 - w_R) \sum_{k=1}^K w_k \pi_c(\psi_* | \xi_k) + w_R \pi_V(\psi_*)$$

Predictive likelihoods and updated weights:

- Marginal likelihoods:

$$m_k = \int_{\Psi} f(Y_* | \psi_*) \pi_c(\psi_* | \xi_k) d\psi_*, \quad m_V = \int_{\Psi} f(Y_* | \psi_*) \pi_V(\psi_*) d\psi_*.$$

- Updated weights: $\tilde{w}_R = \frac{w_R m_V}{w_R m_V + (1 - w_R) \sum w_j m_j}$, $\tilde{w}_k = \frac{w_k m_k}{\sum w_j m_j}$.

- Posterior retains mixture form:

$$(1 - \tilde{w}_R) \sum \tilde{w}_k \pi_c(\psi_* | \xi_k, Y_*) + \tilde{w}_R \pi_V(\psi_* | Y_*).$$

- m_k/m_V acts as Bayes factor; conflict ($m_k \ll m_V$) drives $\tilde{w}_R \rightarrow 1$, automatically discounting historical controls.

- the MAP formulation requires $H \geq 3$ to reliably estimate heterogeneity τ
- When $H = 1$, frequentist random-effects often yield $\hat{\tau}^2 = 0$, underestimating variance uncertainty; moreover, symmetric exchangeability around a mean μ is hard to justify when combining RCT with RWD.

Asymmetric reference model ($\theta_e = \mu$)

Anchors directly to the randomized trial target parameter:

$$\theta_e = \mu, \quad \theta_o \mid \mu, \tau \sim N(\mu, 2\tau^2), \quad k = \sqrt{2}\tau, \quad \tau \sim \text{HN}(s)$$

Conditional on τ , the shrinkage expectation for the trial parameter is:

$$E\{\theta_e \mid \hat{\theta}_e, \hat{\theta}_o, \tau\} = \frac{\sigma_e^{-2} \hat{\theta}_e + (\sigma_o^2 + 2\tau^2)^{-1} \hat{\theta}_o}{\sigma_e^{-2} + (\sigma_o^2 + 2\tau^2)^{-1}}.$$

- Integrates over $\pi(\tau \mid \hat{\theta}_e, \hat{\theta}_o)$, fully propagating heterogeneity uncertainty.
- As conflict increases ($\tau^2 \rightarrow \infty$), observational borrowing vanishes.

Algebraic bridge between shrinkage and power priors

Equating observational shrinkage precision to power prior precision $a_0\sigma_o^{-2}$:

$$a_0\sigma_o^{-2} = \frac{1}{\sigma_o^2 + 2\tau^2} \iff a_0 = \frac{\sigma_o^2}{\sigma_o^2 + 2\tau^2} = \frac{1}{1 + 2\frac{\tau^2}{\sigma_o^2}}$$

Hierarchical variance inflation by $2\tau^2$ is algebraically equivalent to power likelihood discounting (exponentiation) by $a_0 \in [0, 1]$.

Limitations:

- ❶ *Asymptotic normality*: Requires Gaussian summary estimates; fragile for rare-event binary outcomes.
- ❷ *Dispersion vs. bias*: Symmetric offset $N(0, 2\tau^2)$ models random variation, not directional bias; observational data must be rigorously pre-adjusted.
- ❸ *Aggregate evidence*: Operates strictly on contrast summaries, lacking patient-level covariate granularity and heterogeneity effect estimation.

Lin et al. (2026) combine experimental trial (X_e) and observational cohort (X_o) at the individual participant level $X_i = (Y_i, T_i, W_i)$.

Joint power likelihood posterior (two-study setting)

With domain indicator $S_i \in \{e, o\}$ and $\eta \in [0, 1]$:

$$\pi(\phi \mid X_e, X_o, \eta) \propto f_e(X_e; \phi) \{f_o(X_o; \phi)\}^\eta \pi(\phi)$$

Parameter partitioning: $\phi = (\theta, \gamma, \psi) \in \mathbb{R}^p$, $p = p_\theta + p_\gamma + p_\psi$:

- $\theta \in \mathbb{R}^{p_\theta}$: Shared causal treatment effect contrast (parameter of interest).
- $\gamma \in \mathbb{R}^{p_\gamma}$: Trial-specific nuisance parameters.
- $\psi \in \mathbb{R}^{p_\psi}$: Observational nuisance parameters (confounding/selection).

The role of η : Discounting rate ($\eta = 0$: trial only posterior $\pi(\phi \mid X_e)$; $\eta = 1$: full unpenalized pooling) that accommodates both single-arm (control) and two-arm observational data.

Frugal MSM factorisation (Evans & Didelez, 2024)

Partition covariates $W = (C, Z)$ into effect modifiers C and auxiliary Z :

$$f_s(y, t, w) = f_s(t, z | c) f_s(c) \cdot f_{Y(t)|C}^*(y | t, c; \theta) \cdot c_s^*(y, t, z | c; \zeta_s)$$

- f^* models causal CATE $\tau(c)$; θ is variationally independent of nuisance ζ_e ;
- the copula c_s^* links the outcome to $Z|C$ without feedback onto θ .

Causal assumptions:

- 1 *Randomization*: Ensures trial ignorability and positivity, $(Y(t) \perp\!\!\!\perp T | W, S = e)$, $\text{pr}(T = 1 | W, S = e) \in (0, 1)$;
- 2 *Representativeness*: $E\{Y(1) - Y(0) | C = c, S = e\} = \tau(c)$;
- 3 *Contrast transportability*: $E\{Y(1) - Y(0) | C = c, S = o\} = \tau(c)$;
- 4 *Model specification*: Correct model specification for counterfactual $Y(t) | C$.

Cross-validation on experimental trial folds (Lin et al., 2026)

Fit on combined sample $\{D_e^{(-k)}, D_o\}$ with power η , evaluated on held-out trial units $D_e^{(k)}$:

$$\widehat{\text{ELPD}}_{\text{cv}}(\eta) = \sum_{k=1}^K \sum_{i \in D_e^{(k)}} \log \int_{\mathbb{R}^p} f_e(X_i; \phi) \pi(\phi \mid X_e^{(-k)}, X_o, \eta) d\phi$$

Optimal grid tuning: $\hat{\eta} = \arg \max_{\eta \in \mathcal{H}} \widehat{\text{ELPD}}_{\text{cv}}(\eta)$ over a discrete grid $\mathcal{H} \subset [0, 1]$.

Preserving trial internal validity:

- *RCT benchmark*: Evaluating CV strictly on D_e preserves trial validity;
- *Conflict penalty*: Residual confounding or non-transportability in D_o degrades out-of-sample prediction on trial units, forcing $\hat{\eta} \rightarrow 0$;
- *Efficiency gain*: Under causal concordance, variance reduction improves predictive density on held-out trial units, yielding $\hat{\eta} > 0$.

Bernstein–von Mises asymptotic decoupling

As $n_e, n_o \rightarrow \infty$, the joint posterior converges to a Gaussian distribution:

$$\pi(\phi \mid X_e, X_o, \eta) \sim N\left(\hat{\phi}_\eta, (n_e V_e^{-1} + \eta n_o V_o^{-1})^{-1}\right)$$

where $\hat{\phi}_\eta = (n_e V_e^{-1} + \eta n_o V_o^{-1})^{-1}(n_e V_e^{-1} \hat{\phi}_e + \eta n_o V_o^{-1} \hat{\phi}_o)$, with sandwich V_e, V_o .

- *MCMC-free*: Closed-form Gaussian density allows rapid grid search for $\hat{\eta}$, bypassing high-dimensional sampling over large registries.
- *Privacy preservation*: Participating sites can share only summary estimates and sandwich matrices $(\hat{\phi}_s, V_s)$
- Given $\hat{\eta}$, the *ATE* is evaluated via empirical g-computation over trial units: $\hat{\tau}^{(m)} = \frac{1}{n_e} \sum_{i \in D_e} [E\{Y(1) \mid C_i; \theta^{(m)}\} - E\{Y(0) \mid C_i; \theta^{(m)}\}]$, with $\theta^{(m)} \sim \pi(\theta \mid X_e, X_o, \hat{\eta})$.
- **Limitation**: Because $\hat{\eta}$ optimizes joint predictive density over all X rather than $\tau(c)$, moderate confounding may fail to trigger predictive penalties.

Methodological evolution of Bayesian data fusion:

- Transition from static subjective power priors to dynamic, data-adaptive borrowing.
- Pairwise reference shrinkage resolves the single-historical-study ($H = 1$) bottleneck for contrast summaries.
- Combining power likelihoods with frugal parameterization unlocks granular, individual-level borrowing, protecting against observational misspecification.

What's next?

- *Contrast-targeted utility*: Formulating predictive scores focused strictly on the causal contrast $\tau(c)$ to eliminate transient under-coverage.
- *Semiparametric copulas*: Relaxing parametric copula assumptions.
- *Regulatory adoption*: Establishing formal guidelines for dynamic Bayesian borrowing in Phase III rare-disease and pediatric clinical trials.

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Thank you!



Review paper & presentation material

github.com/ClaudiaFran/bayesian-data-fusion-review

Slides revision and $\text{\LaTeX}$ layout design assisted by AI

Methodology	Data level	Cohort setting	Commensurability	Confounding protection
Power prior (<i>Ibrahim & Chen, 2000</i>)	Aggregate / Individual	Single / Multi $H \geq 1$	Fixed elicitation $a_0 \in [0, 1]$	Static precision discounting L^{a_0}
Modified power prior (<i>Duan et al., 2006</i>)	Aggregate / Individual	Multi-study $H \geq 2$	Marginal likelihood Mode $a_0 \rightarrow 1$	Prior-level likelihood discounting
Robust MAP priors (<i>Schmidli et al., 2014</i>)	Summary counts	Multi-study $H \geq 3$	Bayes factor $m_k/m_V \implies \tilde{w}_R \rightarrow 1$	Cromwell's rule mixture with π_V
Pairwise shrinkage (<i>Röver & Friede, 2020</i>)	Summary contrast	Pairwise $H = 1$	Integrated variance τ $a_0 = \frac{\sigma_o^2}{\sigma_o^2 + 2\tau^2}$	Pre-adjustment required (symmetric offset)
Power likelihood + FP (<i>Lin et al., 2026</i>)	Individual	Pairwise or multi $H \geq 1$	Out-of-sample RCT $\text{ELPD}_{\text{cv}} \implies \hat{\eta} \in [0, 1]$	Frugal MSM copula + selective $\hat{\eta} \rightarrow 0$